

Experiment 2: Software Size Measurement and Software Cost Estimation Report

Course Project: Embeddable Cross-Course AI Agent General Architecture Platform
(In Chinese:可嵌入式跨课程 AI Agent 通用架构平台)

1 Experiment 2-1: Software Size Measurement

1.1 Experimental Background

The course project of our group is the “Embeddable Cross-Course AI Agent General Architecture Platform.” The system is designed for teachers and students and covers course management, course resource management, course Agent construction, intelligent Q&A, homework grading, learning analytics, exercise generation, and simulated embedding/adaptation for platforms such as Chaoxing and DingTalk.

Software size measurement is the basis for software engineering management, cost estimation, and economic evaluation. Function Point Analysis measures system size from user-recognizable functions and is suitable for use during the requirements and design stages. To accurately understand the project size, this experiment uses Function Point Analysis.

1.2 Experimental Content

1.2.1 Project Type

Project type: new development project.

This project is developed from scratch. The function point measurement object is all business functionality provided to users at the first delivery of the system. Therefore, this experiment counts all user-recognizable data functions and transactional functions within the system scope.

Within the system boundary: user authentication and authorization, course management, resource management, knowledge point management, Agent construction, intelligent Q&A, homework management and grading, learning analytics, exercise generation, and platform embedding/adaptation.

Outside the system boundary: infrastructure such as PostgreSQL, Redis, and MinIO; external large language model services; and external platform interfaces such as Chaoxing and DingTalk. The infrastructure itself is not counted separately as function points; only the user-recognizable business functions supported by it are counted.

1.2.2 Function Point Identification Method

This experiment mainly adopts the IFPUG Function Point Analysis method and uses the NESMA high-level function point estimation approach for verification. According to the experiment guide, in high-level function point estimation, logical files are counted as low complexity, and transactional functions are counted as average complexity.

Type	Meaning	Complexity Basis in This Experiment	Weight
ILF	Internal logical files maintained within the system boundary	Low	7
EIF	Data maintained by external systems and referenced by this system	Low	5
EI	External inputs that maintain internal data or change system behavior	Average	4
EO	External outputs containing derived data, statistics, or processing logic	Average	5
EQ	External inquiries that mainly display query results without modifying data	Average	4

1.2.3 Data Function Point Statistics

Category	Quantity	Main Identified Objects	Complexity	Function Points
ILF	14	Users/roles, courses, course enrollment relationships, knowledge points/knowledge relationships, course resources, resource slices, Agent templates, Agent instances, Agent workflows, chat sessions/messages, homework, homework submissions, grading results/comments, exercises and learning records	Low	$14 \times 7 = 98$
EIF	5	Chaoxing platform course/user data, DingTalk platform organization/user data, external LLM service configuration and model capabilities, external platform authentication information, external file/message interface data	Low	$5 \times 5 = 25$

1.2.4 Transactional Function Point Statistics

Category	Quantity	Main Identified Objects	Complexity	Function Points
EI	10	Login/registration, course creation and editing, enrollment/member maintenance, resource upload, knowledge point maintenance, Agent configuration and publishing, chat questions, homework creation, homework submission, exercise answer submission	Average	$10 \times 4 = 40$
EO	6	AI answer generation, homework grading reports, class learning analytics reports, individual learning analysis, weak-point alerts, exercise recommendation and generation results	Average	$6 \times 5 = 30$
EQ	3	Course/resource query, chat history query, homework/submission status query	Average	$3 \times 4 = 12$

1.2.5 Unadjusted and Adjusted Function Points

The unadjusted function points are calculated as follows:

$$\text{UFP} = 98 + 25 + 40 + 30 + 12 = 205$$

According to the IFPUG method, the sum of the influence degrees of the 14 general system characteristics is:

$$\sum A_i = 4 + 3 + 3 + 2 + 3 + 4 + 4 + 4 + 4 + 4 + 2 + 3 + 2 + 4 = 46$$

$$\text{VAF} = 0.65 + 0.01 \times 46 = 1.11$$

$$\text{FP} = \text{UFP} \times \text{VAF} = 205 \times 1.11 = 227.55 \approx 228$$

1.3 Experimental Result

The estimated software size of this course project is: Unadjusted Function Points $\text{UFP} = 205$, and Adjusted Function Points $\text{FP} \approx 228$.

1.4 Learning Experiences

Through this experiment, we realized that software size is not equivalent to lines of code. Instead, it should be measured from the perspective of user-recognizable business functions. For a system such as EduAI, which includes a front end, back end, AI Agents, asynchronous tasks, and platform adaptation, clearly defining the system boundary is especially important. Otherwise, infrastructure such as databases, caches, and object storage may be repeatedly counted as part of the software size.

During function point identification, ILF/EIF reflect the data objects managed by the system, while EI/EO/EQ reflect the transactional capabilities of user-system interaction. Only after clarifying the core business objects such as courses, resources, homework, Q&A, grading, learning analytics, and exercises can the subsequent cost estimation have a reliable basis.

1.5 Reflection

The NESMA detailed function point method and the IFPUG method are close in their basic classification and weighting principles. The experiment guide also indicates that the two methods are basically equivalent. When this project is re-examined using the NESMA high-level estimation method, logical files are still treated as low complexity and transactional functions as average complexity, so the resulting UFP remains 205.

The main differences between the two are as follows:

1. Different application purposes. IFPUG provides a complete function point analysis standard and is suitable for precise software size measurement. NESMA inherits the IFPUG method but places more emphasis on engineering practice and proposes simplified methods such as Estimated Function Point Analysis and High-Level Function Point Analysis.
2. Different measurement granularity. IFPUG usually requires detailed identification of indicators such as Data Element Types (DET), Record Element Types (RET), and File Type References (FTR) before determining complexity. NESMA allows high-level estimation when requirements information is incomplete, enabling a quick size estimate with less information.

2 Experiment 2-2: Software Cost Estimation

2.1 Experimental Purpose and Principles

Software size measurement is the basis for software engineering management, cost estimation, and economic evaluation. Function Point Analysis measures system size from user-recognizable functions and is suitable for use during the requirements and design stages.

The purpose of Experiment 2-2 is to select appropriate software labor productivity and person-month rates based on the software size obtained in Experiment 2-1, estimate the project effort, calendar time, and software cost, and verify the result through another cost estimation method.

2.2 Experimental Content

2.2.1 Estimation Basis and Parameters

This experiment uses the software size obtained in Experiment 2-1 as the basic data and refers to the software labor productivity data published by the Beijing Software Cost Evaluation Technology Innovation Alliance. In the discussion record, the selected median software development productivity is 6.96 HH/FP.

The calendar time conversion follows the suggestion in the experiment guide: 22.5 working days per month and 8 hours per day, namely 180 hours per person-month. The person-month rate uses the Shanghai benchmark person-month rate of 31,309 yuan/person-month from the discussion record.

Parameter	Value	Description
UFP	205	Unadjusted function points from Experiment 2-1
FP	228	Adjusted function points from Experiment 2-1, used as the main estimation basis
Productivity	6.96 HH/FP	Median software development productivity
Monthly working hours	180 HH/HM	22.5 working days/month x 8 hours/day
Person-month rate	31,309 yuan/HM	Shanghai benchmark person-month rate
Function point unit price	1,243.52 yuan/FP	Function point unit price method used for verification

2.2.2 Effort and Cost Calculation

The main method first converts function points into effort using productivity, then converts effort into person-months, and finally estimates cost using the person-month rate.

Basis	Size	Effort (HH)	Person-Months (HM)	Software Cost
UFP reference basis	205	$205 \times 6.96 = 1426.80$	$1426.80 / 180 = 7.93$	$7.93 \times 31309 = 248280.37$ yuan, approximately 248,300 yuan
FP main basis	228	$228 \times 6.96 = 1586.88$	$1586.88 / 180 = 8.82$	$8.82 \times 31309 = 276145.38$ yuan, approximately 276,100 yuan

In the table, $180\text{hours} = 22.5\text{ days} \times 8\text{hours/day}$.

Therefore, using the adjusted function points FP as the main basis, the estimated effort of this project is approximately 1,586.88 person-hours, or 8.82 person-months. The estimated software development cost is approximately 276,100 yuan.

2.2.3 Cost Result Verification

The verification method adopts the direct estimation method of “Function Points x Function Point Unit Price.” The function point unit price is 1,243.52 yuan/function point.

Basis	Verification Cost	Difference from Main Method	Difference Rate
UFP = 205	$205 \times 1243.52 = 254921.60$ yuan, approximately 254,900	$254921.60 - 248280.37 = 6641.23$ yuan	Approximately 2.67%

	yuan		
FP = 228	228 x 1243.52 = 283522.56 yuan, approximately 283,500 yuan	283522.56 - 276145.38 = 7377.18 yuan	Approximately 2.67%

The difference between the two methods is approximately 2.67%, which is significantly lower than 20%. This indicates that the current cost estimation result is relatively stable.

2.2.4 Reflection

If the difference between the two methods exceeds 20% in an actual project, the size boundary, function point classification, complexity judgment, productivity value, and rate source should be reviewed item by item. If necessary, multiple benchmark parameter sets should be used for interval estimation, and the budget should be adjusted by considering project risk reserves.

2.3 Experimental Conclusion

Using FP = 228 as the main basis and the median productivity of 6.96 HH/FP, the project effort is estimated to be approximately 1,586.88 person-hours, equivalent to 8.82 person-months. At a rate of 31,309 yuan/person-month, the software cost is approximately 276,145.38 yuan, or about 276,100 yuan.

After verification using the function point unit price method, the estimated cost under the FP basis is approximately 283,500 yuan, with a difference of about 2.67% from the main method. Since this does not exceed the significant difference threshold of 20%, the estimation result can be used as a reference for the size and cost planning of the course project.

2.4 Learning Experiences

The cost estimation section shows that software cost depends not only on the number of function points, but also on productivity, person-month rate, and non-functional characteristics. By comparing the main method with the verification method, we can see that the estimation difference in this project is small. However, in actual development, reasonable risk buffers should still be set according to team experience, requirement changes, AI service invocation risks, and platform integration difficulty.

References

1. Experiment Instruction + Exp 2-1 Software Size Measurement.
2. Experiment Instruction + Exp 2-2 Software Cost Estimation.
3. Beijing Software Cost Evaluation Technology Innovation Alliance software labor productivity and cost benchmark data.
4. Project documents: README, architecture document, and front-end design document of the Embeddable Cross-Course AI Agent General Architecture Platform.